Heatmaps

Page Summary

The `heatmapTiles` (/maps/documentation/air-quality/reference/rest/v1/mapTypes.heatmapTiles/lookupHeatmapTile) endpoint provides air quality related heatmaps through a tile overlay. A tile overlay is a collection of PNG image tiles that can be displayed on top of a Google Maps. Each image tile is (256 x 256 pixels).

About heatmap tiles

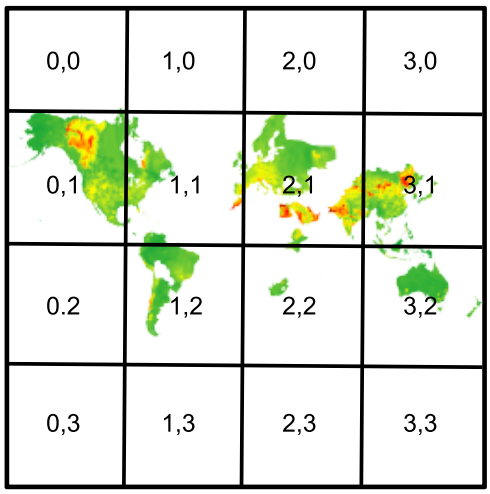
The tile overlay is a grid assembled from a collection of tiles, each assigned (`X`,`Y`) coordinates. The tile with coordinates (0,0) is always at the northwest corner of the map. The `X` values increase from west to east, while the `Y` values increase from north to south.

The **`zoom level`** determines how large or small the contents of a map appear in a map view. The tile grid size is determined by scaling the `X` and `Y` coordinates exponentially by the zoom level.

$$\text{gridSizeX} = X^{\text{zoomLevel}}$$
$$\text{gridSizeY} = Y^{\text{zoomLevel}}$$

For example, at a zoom level of 2, the world map is represented using a 4 x 4 matrix, for a total of 16 tiles. The zoom also determines what is the max value allowed for the coordinates.

The following image shows a `US_AQI` heatmap with a zoom level of 2, along with the coordinates of each tile:



See Map and Tile coordinates (/maps/documentation/javascript/coordinates) for additional details.

About the heatmap endpoint

A heatmap tile is represented by a bytes array containing the tile data as a PNG image. You request current air quality heatmap tiles using the `heatmapTiles` (/maps/documentation/air-quality/reference/rest/v1/mapTypes.heatmapTiles/lookupHeatmapTile) endpoint by sending an HTTP GET request to:

```
https://airquality.googleapis.com/v1/mapTypes/TYPE/heatmapTiles/Z/X/Y?key=YOU
```

All request parameters are required in a request:

- `TYPE` - The type heatmap to return. See allowed values (/maps/documentation/air-quality/reference/rest/v1/mapTypes.heatmapTiles/lookupHeatmapTile#maptype) .
- `Z` - Zoom level that determines how large or small the contents of a map appear in a map view. Allowed values are 0-16, where a value of 0 displays the entire world in a single tile.
- `X`,`Y` - The coordinates of the tile to retrieve, relative to the north west corner (0,0). `X` values are increasing from west to east and `Y` values are increasing from north to south.

Tile coordinates have to be valid for the specified zoom level. For example, if you set the zoom level to 2, and request a tile at coordinates of 10,10, the API returns an error.

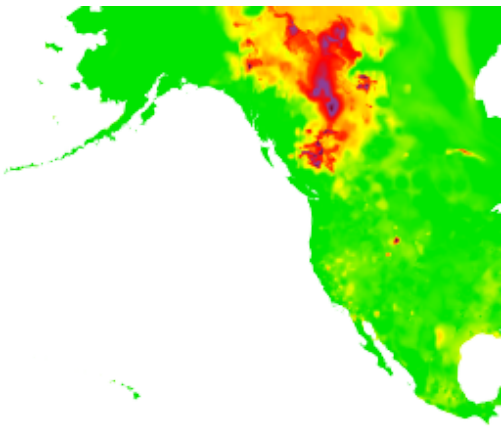
- `YOUR_API_KEY` - Your application's API key. This key identifies your application for purposes of quota management. For more information, see get a key (/maps/documentation/air-quality/get-api-key).

Example heatmap request

The image above shows a `US_AQI` heatmap with a zoom level of 2, along with the coordinates of each tile. The following example uses the `heatmapTiles` endpoint to request the tile at coordinates 0,1 from the heatmap:

```
https://airquality.googleapis.com/v1/mapTypes/US_AQI/heatmapTiles/2/0/1?key=YO
```

The tile appears as:



For more information on overlaying tiles on top of the default map, see Overlay Map Types (/maps/documentation/javascript/examples/maptype-overlay).

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